UNIVERSITY OF CENTRAL PUNJAB

(Faculty of ITCS)

Operating Systems (OS)

Assignment No 2 **Date: 04 Nov 2025**

Total Marks: 100 **Due Date: 11 Nov 2025**

Assignment Guidelines:

*Assignment is to be done in **groups** of **02 students** each.*
*Assignment is to have a **Title page : Course Name**, **Assignment No.**, **Student Name**, **Reg No.**, **Section**.*
*Program Code to be submitted in **hard copy**; should be in small font for readability and indented/ well commented.*
***Screen shots** of code functionality are to be provided.*
*There will be a **quiz/viva** based on assignment on the submission day; also, **demo** of running code.*
Late submission will cause you to lose 25% per day*; after three days of deadline no submissions will be entertained.*

CLO 2 Questions

1. Write a program that creates two child processes and uses **Pipes** to pass a message in a chain: Parent → Child1 → Child2. Show that program runs correctly by displaying relevant outputs in terminal windows during program execution.
2. Create a simple client-server application having full duplex communication between them by using **Anonymous pipes**. The client (child) sends a command to server (parent), and server responds accordingly. For example, client sends a command to open a file file1.txt to server, server opens the file, reads first 10 lines of file, and then sends these lines to client. Client saves the received data to another file, file2.txt.
3. Build a simple chat application for full duplex communication using two **Named pipes** (one for each direction of communication). Provide screen shots of message communication for at least 4 message exchanges.
4. Implement a concurrent file-processing system:
   - Parent reads a line from a file
   - Sends the line to a Child via a **Pipe**
   - Child converts the line to uppercase and writes it to **FIFO** pipe
   - Parent reads the result from **FIFO** and prints it on display

_____ End _____